

Designing a Multimodal Robot Pet for Older Adults by Young Adults

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Abstract

Social robots that promote mental well-being have been developed for healthy aging. Negative attitudes of people, both young and old, towards such robots currently affect their acceptance. Involving young adult relatives in the design of a social robot with sound/voice responses to multimodal touch, gesture, emotion and speech inputs that can be given as a gift to an elderly family member to improve the chances of its acceptance has not been studied. To this end, we conducted two user interaction design studies, the Identification and Matching studies, with 20 and 21 young adults, respectively, and measured expected changes in attitudes using the Negative Attitudes Towards Robots Scale (NARS) and the General Attitudes Towards Robots Scale (GAToRS) questionnaires. Our results show that negative attitudes evolve positively following the relatives-designed approach, which could promote a positive elderly-robot interaction and increasing social robot acceptance.

CCS Concepts

• **Human-centered computing** → **User centered design; User studies.**

Keywords

Social Robots, Older Adults, Young Adults, Negative Attitudes Towards Robots, Robot Acceptance, Relatives-Designed Robot, Elderly-Robot Interaction

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1 Introduction

Social robots are defined as autonomous machines that possess social intelligence to interact with people in socially acceptable ways [23]. These robots often embody human or animal characteristics and interactions, and are designed to support the well-being of older adults [62, 85]. Several social robots have demonstrated the potential to serve older adults as therapists [20], coaches [25], social facilitators [58], assistants [70], monitors [22], and companions [21]. Social robots are emerging as an effective tool for improving the mental well-being of older adults by providing emotional support and increasing social interaction [12, 62]. Despite the potential of social robots, some robots have recently caused negative attitudes toward social robots among older adults [80]. Previous research has shown that people both young and old may have negative attitudes toward robots, which currently affect their acceptance [6, 16], as these different age groups generally reported similar negative attitudes toward robots [4]. In this paper, we explored this issue by designing with young people for their older adult relatives.

We considered social robots that provide companionship, or companion robots, which have been co-designed by involving a broader group of stakeholders in the design process, such as the elderly person, caregivers, medical professionals, psychologists, and family members [14, 28, 31, 35, 66, 72, 73]. This approach could provide intergenerational social connectedness [63, 69, 87] by involving different generations, such as older adults with children [36] or young adults [54], and promote mental well-being. However, some older adults may face cognitive and communication challenges, such as when people with dementia have difficulty remembering activity instructions, experience stress when using unfamiliar technology, and are unable to express themselves or their needs [74, 78, 84]. This prevents them from fully participating in social interaction, understanding conventional design activities, and benefiting from their experience [33, 42].

On the other hand, giving gifts to older adults in the form of expressing gratitude has contributed to improving their overall well-being [18, 71]. Maintaining relationships between family members of young and older adults through technology could help create, recreate, and retrieve memories [17, 24, 83]. Consequently, involving young adults in the design process, excluding older adults, with the goal of giving them a relatives-designed and personalized robot as a gift, as a form of expressing gratitude [67], and in order to playfully create, recreate, and retrieve personal memories with them, has not been studied before [18, 71]. This approach of developing relatives-designed robots for the purpose of gift-giving could surprise and excite older adults, thereby improving their negative attitudes toward social robots.

Previous pet robots, such as PARO, AIBO, and JoyForAll, were designed to make only animal-like sounds. However, in the literature, older adults preferred interactive and playful features in robotic pets so that they could communicate and interact with the robot as a play partner through verbal conversations [5, 10, 34]. To address this gap and to better understand the attitudes of older and young adults toward social robots, we conducted two user studies with young adults: (1) an interaction *identification* study with 20 young adults [2]; and (2) an interaction *matching* study with 21 young adults, to design sound/voice responses to a set of multimodal touch, gesture, emotion, and speech inputs for a talking pet robot for their older adult relatives. We measured the expected changes in older adult relatives' attitudes using the Negative Attitudes Towards Robots Scale (NARS) and the General Attitudes Towards Robots Scale (GAToRS) questionnaires before and after the interaction design activities.

2 Related Work

Social robots are becoming prevalent, which demands an understanding of the factors that affect their success and users' attitudes towards these robots [80]. Previous research has suggested that several factors may underlie these attitudes, including their familiarity, appearance, usability, and emotional attachment [16, 32, 77]. Familiarity with social robots could contribute to more positive attitudes towards robots compared to people who are less familiar with these robots [77]. The physical appearance of a robot is closely related to the users' attitudes towards robots, where an appealing appearance would create positive attitudes and an enjoyable interaction experience [39, 80]. In addition, the usability of such robots could affect older adults' attitudes towards them, with a proportional relationship between the robot usability and acceptance, resulting in overall positive attitudes towards these robots [16]. Users' emotional attachment to social robots is another important factor in shaping their attitudes towards these robots, as users may see these robots as more than just machines, and may feel companionship and connection towards them [5]. These factors could uncover people's attitudes towards robots, which could inform researchers and designers about the development and design of social robots that could affect users' attitudes.

Various participatory design methods have been applied to the design of social robots, with a focus on designing social robots for and with older adults. This approach has led to methodological innovations for an equal cooperative partnership between young and

older adults [53, 61, 69]. The design of social robots revolves around the application of the user-centered design (UCD) approach [51], which incorporates various methodologies, including a co-design approach. Rogge's review suggests that the co-design of social robots, with a focus on social integration, effective communication, and UCD, can lead to positive outcomes and acceptance among users, especially older adults [68]. By exploring potential solutions and future designs, co-design for and with older adults has been conducted to support the development of robots [45, 47, 52, 57, 60]. The existing literature has emphasized intergenerational collaboration in the design of social robots through various open-ended evolution methods, which could promote the well-being of older adults and intergenerational relationships [11, 54, 59, 79]. In the literature, robots have been designed with intergenerational groups considering the inputs and outputs of the robot interactions. An elicitation method was used to elicit users' preferred design features, demonstrating the importance of UCD in the selection and acceptance of the desired robot [9, 59]. Previous work mainly helps us to understand the design of robot features for and with older adults. However, it is essential to understand older adults' attitudes towards social robots and robots designed by their relatives, taking into account different types of multimodal interactions.

3 Methodology

Before proceeding with our user studies, we obtained informed consent from participants according to our Institutional Review Board (IRB-010423/6280) approved procedure. All participants were recruited through physical and digital advertisements. Participants were free to withdraw at any time during the studies.

3.1 Study 1: Identification

3.1.1 Apparatus and Task Design. Our study was conducted in a user experience (UX) laboratory on our university campus. The apparatus consisted of JoyForAll cat and a dog robots hidden in their packaging boxes, a 45-inch PC monitor displaying a QR code for initial survey and demographic data collection, and a MacBook laptop for session recording and data collection (see Fig. 1). We used JoyForAll robots as probes because some older adults prefer animal-like robots for emotional well-being, featuring soft fur, medium size, and realistic interactions [1, 8]. In addition, these robots featured low cost, making them affordable, and their design mimicked real pet, making them more familiar and realistic and more preferable than other unfamiliar robots, i.e., Paro [8, 41]. To design our tasks, we generated a list of four modes of multimodal input derived from the literature, i.e., touch [15, 27, 30, 37, 48, 76, 88], emotion [3, 26, 46, 75, 89], gesture [7, 40, 44, 49], and speech [38, 50, 55]. We compiled 42 multimodal interaction inputs corresponding to the four modes, focusing on inputs that were mentioned in at least two publications in the literature, in order to identify the corresponding robot sound/voice responses by participants. Specifically, we adapted the list of inputs to fit interaction with a talking social pet robot by including 16 hand-based touch inputs, 7 emotions, 10 gestures, and 9 speech interaction inputs (see Fig. 2).

3.1.2 Participants and Procedure. We recruited 20 young adults from our university campus. We stopped recruitment after 20 participants had enrolled, anticipating data saturation as suggested

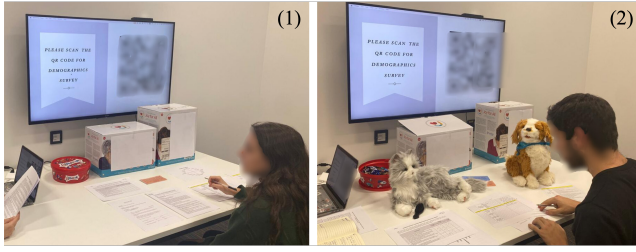


Figure 1: A participant is answering the NARS questionnaire (1) before interacting with the JoyForAll robots hidden in their packaging boxes and (2) after completing the interaction identification tasks with the JoyForAll robots.

in the literature [13, 29, 82]. The study lasted approximately 45 minutes and consisted of several phases.

Introduction (10 minutes): We started with an introduction phase where we explained the purpose of the study, i.e., the potential challenges that older adults may face, such as the need for companionship and emotional support. We also explained the tasks and the procedure of the study. Participants then read the information sheet and familiarized themselves with the apparatus and the tasks. They then signed the informed consent form, scanned a QR code on the PC monitor, and provided demographic information about themselves and their elderly family members. The demographics of the participants are: 12F, 8M, aged 24–30 years with a mean and SD age of 27.4 ± 5.8 years. The demographics of an elderly relative for whom the participants chose to design are: 13F, 7M, aged 75–84 years with a mean and SD of 76.9 ± 6.7 years, with nine participants designing for their grandmothers, four for their grandfathers, two for their mothers, three for their fathers, and two for their great aunts. Participants also answered the following questions: (1) Is the older adult a pet person? If yes, then what type of pet? 13 out of 20 young adults said that the older adult relative they were designing for was a pet person, and they owned several different types of pets, including cats, dogs, sheep, birds, horses, and camels. (2) Do you have previous experience with robots? If yes, then could you please specify the type of robot? Seven of the twenty young adults said they had previous experience with robots – three with Arduino-based bots, two with Nao, one with a vacuum cleaner, and one with a manufacturing robot. (3) Have you seen the Negative Attitudes Towards Robots Scale (NARS) before? None of the 20 participants had seen the NARS questionnaire before.

NARS Questionnaire Before Interaction (5 minutes): Participants then completed a NARS questionnaire before interacting with the pet robots. We asked them to imagine their elderly family members’ (negative) attitudes toward social robots while completing the questionnaire. This captured older adults’ current expected negative attitudes toward social robots from the perspective of their young adult relatives.

User-robot Interaction (Passive (5 minutes)): Participants then unboxed the JoyForAll robots, one package at a time, and interacted with the robot in a passive (power off) state. We had covered the photo of the robots on the box with a piece of paper, so that the participants were not biased beforehand.

Interaction Identification (15 minutes): Participants then took on the role of an interaction designer following Ali et al.’s user identification study [2]. They identified sound/voice responses for a social pet robot, considering 42 multimodal (including touch, emotion, gesture, and speech modes) inputs in a randomized order. They were free to design a pet, human, or any other type of sound or voice response. We asked participants not to repeat a response within each mode. In the end, participants designed 840 (20×42) sound/voice responses.

User-robot Interaction (Active (5 minutes)): Participants then interacted with the JoyForAll robots one at a time, but this time in the active (on) state, and then imagined how their elderly family members would react and perceive these robots as they interacted with them.

NARS Questionnaire After Interaction (5 minutes): Participants then re-imagined their elderly family members’ (negative) attitudes towards the social pet robots they had designed to interact with, and completed the NARS questionnaire after interacting with the JoyForAll pet robots.

3.1.3 Findings. 1. Agreement Rate (AR): We performed the AR analysis, which characterizes the degree of consensus between participants’ suggestions for the robot’s voice/sound responses to different inputs. We started by classifying the robot sound/voice responses collected from participants and used a process called “binning” [90] to combine the similar outputs; for example, we classified (“what I did” similar to “what I did”). As a result, we ended up with 344 unique sound/voice responses. We then applied statistical analysis to these responses using the AR computation [81, 86] along with AGATe (AGreement Analysis Toolkit) to assess the participants’ agreement on the 344 unique sound/voice responses for the 42 multimodal interaction inputs across touch, emotion, gesture, and speech modes. We present the level of agreement of the robots’ sound/voice responses for the 42 multimodal inputs in Fig. 2, where the inter-participant response agreement level is interpreted as low $AR \leq 0.1$, medium $0.1 < AR < 0.3$, high $0.3 \leq AR \leq 0.5$, and very high $AR > 0.5$. In our study, out of 42 inputs, 14 (33%) have low agreement, 18 (43%) have medium agreement, 6 (14%) have high agreement, and 4 (10%) have very high agreement. Ten interactions reached consensus because they had a high or very high level of agreement. Overall, the AR of our study ranges from 0.03 to 0.81.

2. NARS Before and After Robot Interaction: We obtained quantitative data using the NARS questionnaire [56]. Participants rated the NARS questionnaire on a 5-point Likert scale, which captures the expected negative attitudes toward social robots from the perspective of young adult designers while imagining the reactions of their elderly relatives. We analyzed the NARS questionnaire according to its three subscales, S1–situations of interaction with robots, S2–social influence of robots, and S3–emotions in interaction with robots. To display the NARS scores in a similar way, i.e. a lower score means less negative attitude, we show the S3 subscale inverted. This means that a lower value of S3-inverted indicates that participants had a more positive attitude towards robots (see Fig. 3 and Table 1).

We tested whether there was a significant difference in the NARS scores before and after the robot interaction design session across

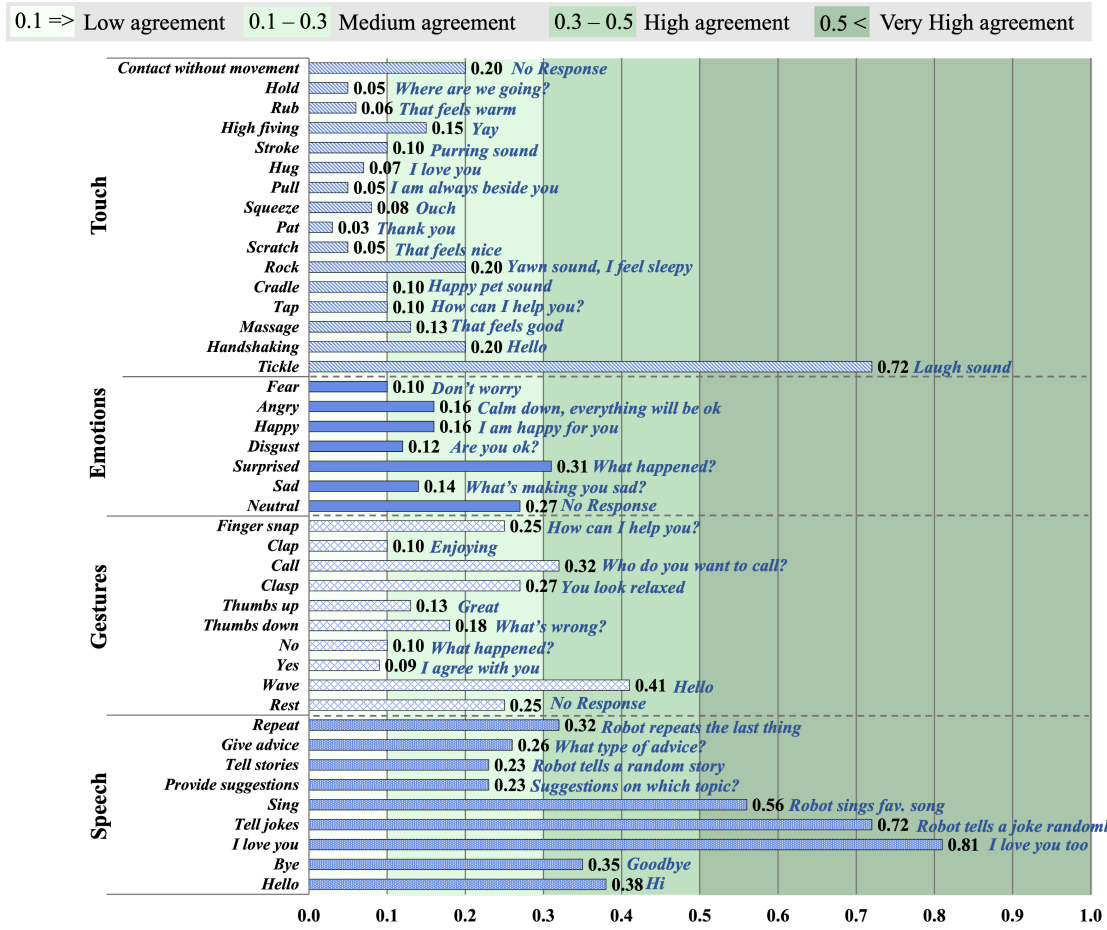


Figure 2: The agreement rates of the identification study representing all 42 user inputs or actions under the four modes – touch, emotion, gesture and speech, are shown. The agreed robot’s responses or outputs for the inputs are shown on the right.

Table 1: NARS scores before and after design interaction session for the three subscales: S1–Situations of interaction, S2–Social influence, and S3–Emotions in interaction (Inverted)

	Before		After	
Scale	(Mean ± SD Median)		(Mean ± SD Median)	
S1	3.02 ± 1.11	3.00	2.04 ± 0.90	2.00
S2	3.67 ± 1.04	4.00	2.59 ± 1.12	2.00
S3	3.30 ± 1.06	3.00	1.77 ± 0.65	2.00

the S1–situations of interaction with robots, S2–social influence of robots, and S3–emotions in interaction with robots. We performed a nonparametric Wilcoxon signed-rank test [64] and found significant differences with ($Z = -6.8$, $p < 0.001$, $r = -0.8$) for S1, ($Z = -6.1$, $p < 0.001$, $r = -0.7$) for S2, and ($Z = -6.1$, $p < 0.001$, $r = -0.9$) for S3–inverted, where a negative Z –value indicates that the scores after the interaction design session are much lower than before, showing a positive shift in the scores and a less expected negative attitudes toward robots.

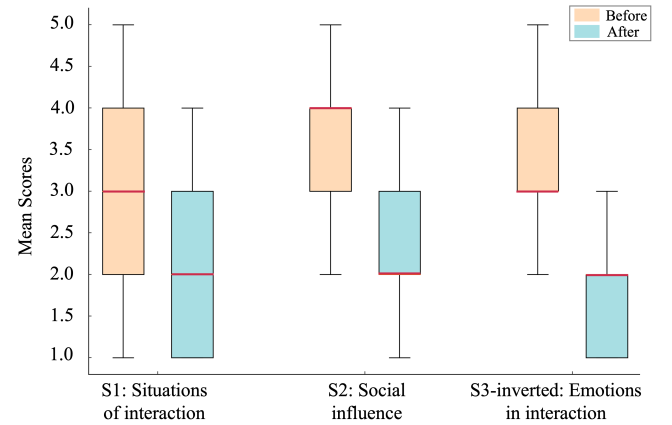


Figure 3: NARS questionnaire scores for each subscale (S3 inverted) before and after interaction design for a talking pet robot.

3.2 Study 2: Matching

3.2.1 Apparatus and Task Design. We conducted the matching study in a social robotics lab on the university campus, using the same apparatus as for the identification study (see Fig. 1). To design our task, we generated a randomized worksheet with two lists of the same 42 multimodal inputs of the four modes across the agreed sound/voice responses obtained in the identification study, as shown in Fig. 2.

3.2.2 Participants and Procedure. In the matching study, we recruited 21 new young adults from our university campus who took 45 minutes to complete the entire study. The study procedure follows the same structure as the identification study, but with some refined sections that are aligned with the goal of our matching study.

Introduction (10 minutes): Participants read the information sheet, signed the consent form, and scanned a QR code to provide demographic information about themselves and their elderly family member. The participant demographics are: 12F, 9M, aged 24–30 years with a mean and SD age of 28.0 ± 3.5 years. The demographics of their elderly family members are: 16F, 5M, aged 65–74 years with a mean and SD age of 71.6 ± 9.3 years. Ten participants designed for their grandmothers, two for their grandfathers, three for their fathers, three for their mothers, one for their mother-in-law, and two for their great aunts. The participants answered the same questions as in the identification study, except for the last question, where the NARS questionnaire was replaced by the General Attitudes Towards Robots Scale (GAToRS) questionnaire in order to capture the negative and positive attitudes towards the robots. We obtained the following results: (1) Eleven out of 21 young adults said that the older adults they would design for were pet people, where they owned different types of pets, including cats, dogs, birds, and sheep. (2) Five of the 21 young adults said they had previous experience with robots, including two with robot vacuum cleaners, two with a social robot (Pepper), and one with a 3D printing robot. (3) None of the 21 participants had seen the GAToRS questionnaire before.

GAToRS Questionnaire Before Interaction (5 minutes): Participants were then asked to imagine their elderly relatives' (positive and negative) attitudes towards social robots and to complete a GAToRS questionnaire before interacting with the pet robots.

User-robot Interaction (Passive (5 minutes)): Participants then unboxed the covered JoyForAll robots, one package at a time, and interacted with the robot in a passive (power off) state.

Interaction Matching (15 minutes): Participants then took on the role of an interaction designer to match between two randomized lists of 42 multimodal (i.e., the touch, emotion, gesture, and speech modes) inputs with the agreed-upon sound/voice responses for a social pet robot. We asked participants not to repeat any responses within each mode and to match the outputs only once uniquely within each mode. They could match any output by repeating it across different modes (e.g., Gesture: 'Finger snap' and Touch: 'Tap' could be matched with the output "How can I help you?"). This ensured that all 42 inputs were matched across different modes.

User-robot Interaction (Active (5 minutes)): Participants then interacted with the JoyForAll robots again, this time in an active (switched on) state. We asked them to consider how their elderly

family members would react to and perceive these robots after active human-robot interaction.

GAToRS Questionnaire After Interaction (5 minutes): Participants then completed the GAToRS questionnaire after interacting with the JoyForAll pet robots, by imagining their elderly relatives' (positive and negative) attitudes toward the social pet robots they had designed for them.

3.2.3 Findings. 1. Agreement Rate: We analyzed our results from the matching study by following the same protocol as the identification study to calculate the agreement rate. We evaluated the participants' agreement on the sound/voice responses mapped to the 42 multimodal interaction inputs across touch, emotion, gesture, and speech modes. We present the updated agreed robot sound/voice responses from the identification study based on the final results of the matching study (see Fig. 4), along with the updated mapped sound/voice responses for the 42 multimodal inputs. In the matching study, 13 (31%) of the 42 inputs have a low agreement ($AR \leq 0.1$), 16 (38%) have a medium agreement ($0.1 < AR < 0.3$), 3 (7%) have a high agreement ($0.3 \leq AR \leq 0.5$), and 10 (24%) have a very high agreement ($AR > 0.5$). 13 interactions reached consensus because they had a high or very high level of agreement. Interestingly, 10 of them belonged to the speech mode, which means that all of the speech inputs had a very high AR. The speech mode tended to have higher agreement among participants than the other modes, perhaps because the participants found it easier to map the voice (speech) output to the corresponding speech input due to the intuitive and natural nature of the conversation.

2. GAToRS Before and After Robot Interaction: We obtained quantitative data using the GAToRS questionnaire [43]. Participants rated the GAToRS questionnaire on a 7-point Likert scale, capturing the expected positive and negative attitudes toward social robots of young adults imagining the reactions of their elderly relatives. The positive and negative subscales in the GAToRS questionnaire indicate both personal and societal levels with positive (P+, S+) and negative (P-, S-). We tested whether there was a significant difference in the GAToRS scores before and after the robot interaction design session across the personal and societal levels with both the positive and negative subscales. We performed a nonparametric Wilcoxon signed-rank test and found significant differences with ($Z = -3.6, p < 0.01, r = -0.79$) for P+, ($Z = -3.6, p < 0.01, r = -0.70$) for P-, ($Z = -3.5, p < 0.01, r = -0.77$) for S+, and ($Z = -2.8, p < 0.01, r = -0.60$) for S-, indicating a positive shift in the expected attitudes toward robots for all four subscales as shown in Table 2 with the mean, SD and median. Fig. 5 also shows the mean scores of GAToRS before and after for all the four subscales, where higher scores indicate more positive or negative attitudes towards robots.

4 Discussion

Our identification and matching studies of the design of a personalized talking pet robot showed that the key design of the robot's agreed sound/voice responses to multimodal inputs contributed to less negative and more positive attitudes toward the robot in older adults as imagined by their younger relatives. The preference for choosing a cat or a dog as a pet for their elderly relative was not balanced across the participants. In the identification study, most participants chose – cat (14), dog (2), and cat/dog (4), whereas in the

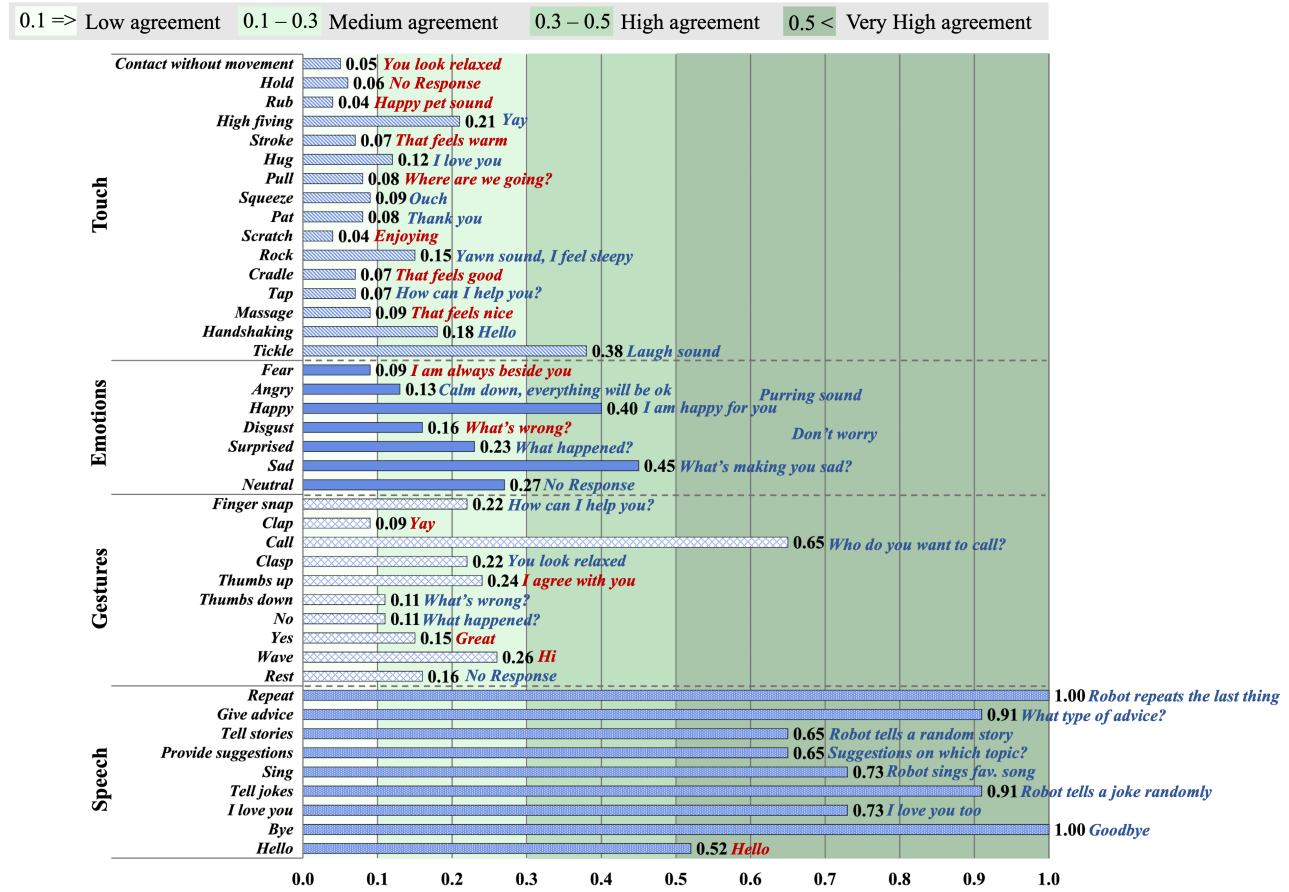


Figure 4: The agreement rates of the matching study representing all 42 user inputs or actions under the four modes – touch, emotion, gesture and speech, are shown. The agreed robot's responses or outputs are shown on the right

Table 2: GAToRS scores before & after design interaction sessions Personal Level Positive Attitude (P+), Personal Level Negative Attitude (P-), Societal Level Positive Attitude (S+), and Societal Level Negative Attitude (S-)

	Before		After	
Scale	(Mean ± SD Median)		(Mean ± SD Median)	
P+	4.14 ± 0.40 3.95		5.28 ± 0.25 5.24	
P-	3.55 ± 0.70 3.86		2.64 ± 0.58 2.52	
S+	4.94 ± 0.13 4.90		5.73 ± 0.38 5.71	
S-	5.41 ± 0.32 5.57		4.73 ± 0.43 4.86	

matching study, participants chose – cat (9), dog (5), and cat/dog (7). Participants unboxed both robots and were then free to consider either or both robots to design/identify/match the interactions. Results may be biased toward a robot cat due to personal preference for a pet type. The sample size of our studies was relatively small, but our findings with the attitude questionnaires are informative. In the future, separate identification or matching studies could be conducted to inform the interaction design for specific pet robots with

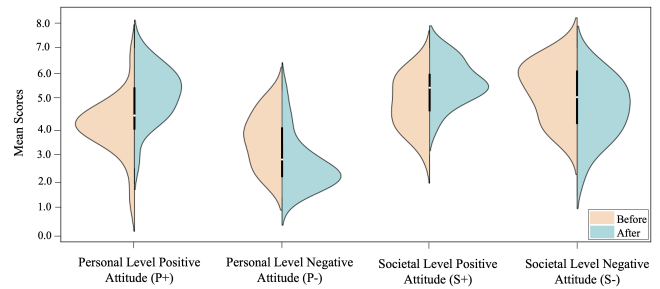


Figure 5: GAToRS questionnaire scores for the four subscales– Personal Level Positive Attitude (P+), Personal Level Negative Attitude (P-), Societal Level Positive Attitude (S+), Societal Level Negative Attitude (S-) before and after interaction design for a talking pet robot.

a larger sample of participants. A longitudinal iterative study with real-world trials could also be conducted in the future to improve the validity of the results.

In some developed countries, older adults prefer more functional and human-like robots [19]. Participants in our study indicated that this may be due to exposure to cinematic depictions of robots that

are more human-like and mostly positive attitudes toward them [65]. More recently, animal-like robots are finding their place in society as therapeutic (PARO), educational (Bee-Bot) and companion (AIBO) robots. However, people's exposure to these robots is limited without popular portrayals on television and in the movies, which may change in the future for social pet robots, leading to their wider acceptance and impact.

5 Conclusion

In this paper, we investigated the negative attitudes of older adults toward a social robot pet from the perspective of their younger relatives when designing the robot's sound/voice responses to multimodal touch, emotion, gesture, and speed inputs for playful interaction. We measured attitudes during the design of the robot's responses before and after the interaction design activities through identification and matching studies using the Negative Attitudes Towards Robots Scale (NARS) and the General Attitudes Towards Robots Scale (GAToRS) questionnaires. The identification study, in which participants proposed different sound/voice responses to 42 multimodal inputs, showed 24% consensus in the proposals. The matching study, in which participants matched the agreed-upon sound/voice responses from the identification study to the 42 multimodal inputs, showed 31% consensus in mapping the multimodal inputs to the sound/voice outputs. The speech mode achieved the complete and highest consensus among the modes. The results of the NARS and GAToRS questionnaires before and after the user interaction with the robots showed a positive shift, indicating a decrease in the negative attitudes of older adults toward robots from the perspective of their younger relatives. We hope that our research will contribute to the field of human-robot interaction design and future robot designs that would foster meaningful connections between young adults and their older family members.

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